

CMM Stage 1c — strict $h=0$ co-area mesh forward operator

Test cell: $(\tau, \gamma) = (2.0, 0.098)$, $G = 21^3$, $h_{\text{kernel}} = 0.45\sqrt{\Delta u} = 0.285$

Fixed point P^* : emin15-certified longdouble FP ($\|\Phi_{\text{kernel}}(P^*) - P^*\|_{\infty} = 4.00e - 15$)

CMM forward operator: surfaces extracted by marching cubes at N_{bins} price levels, sliced by hyperplanes $u_k = U$ (arc-length integration; $h \equiv 0$).

CMM ($n_{\text{bins}}=128$) cube available.

$\|\Phi_{\text{CMM}} - \Phi_{\text{kernel}}\|_{\infty}$ on whole cube: 3.300e-01

median: 6.230e-03

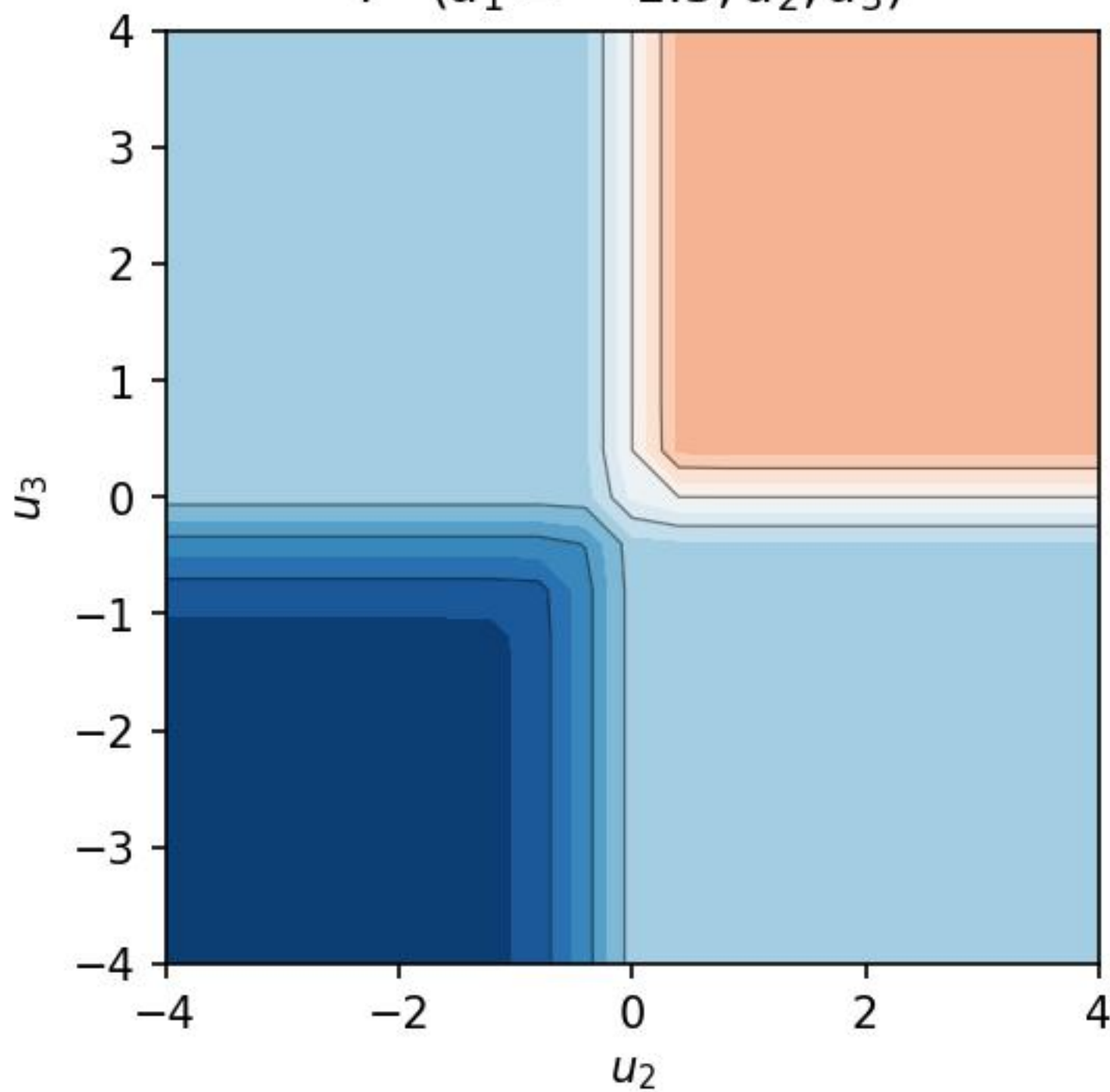
p90: 2.995e-01

Interpretation: the cell-wise discrepancy IS the $O(h)$ error of the kernel operator, varied locally (small in flat regions, large near Morse-critical price levels). Median is $O(h^2)$, tails reach $O(h)$.

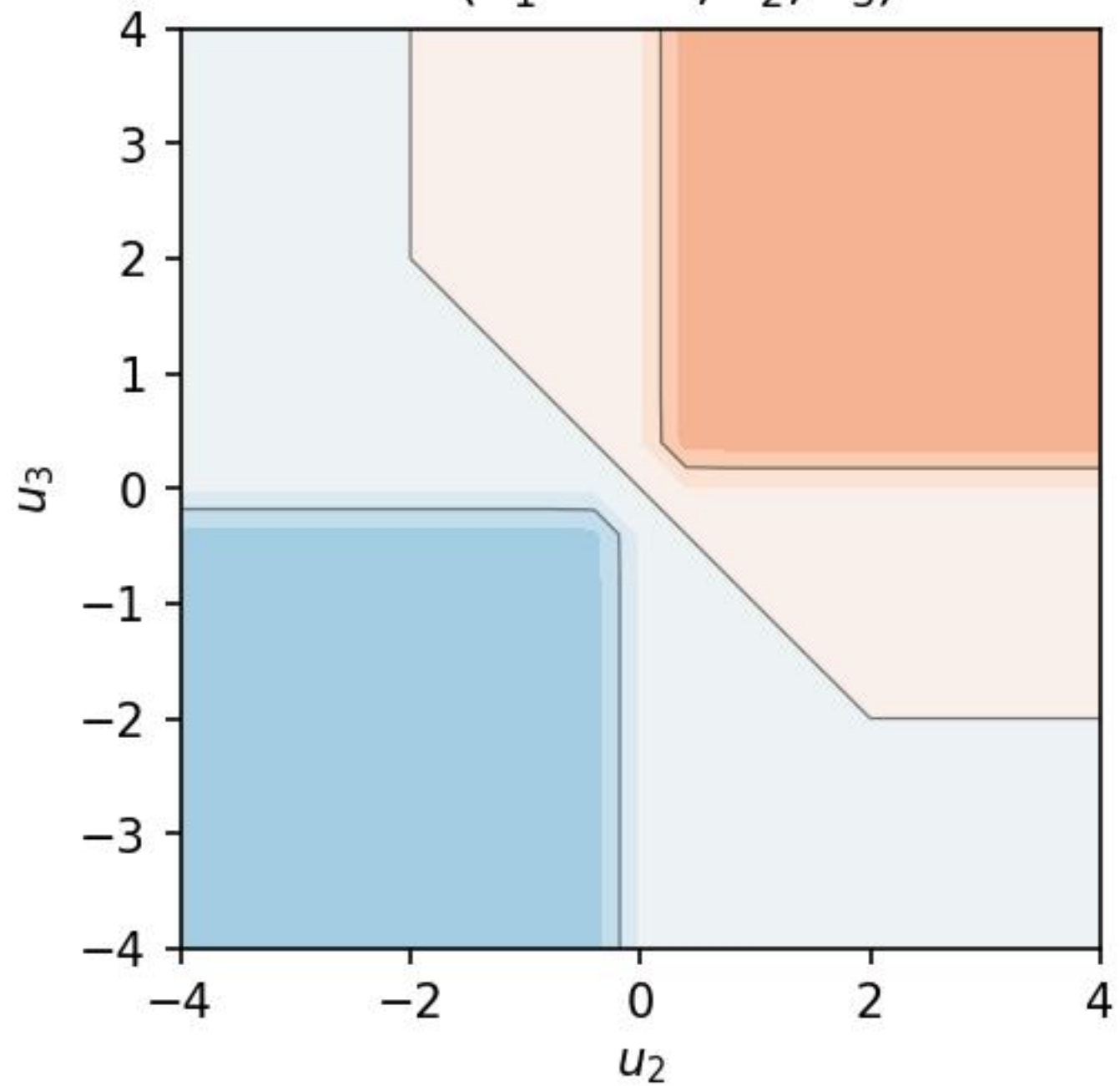
Pages 2-5 show: P^* slices; Φ_{CMM} output slices; $|\Phi_{\text{CMM}} - \Phi_{\text{kernel}}|$ error map; and the extracted level curves $\{P^* = p\}$ in 2D --- those are exactly the curves CMM integrates over.

Page 2 — Fixed point $P^*(u_1, u_2, u_3)$ slices

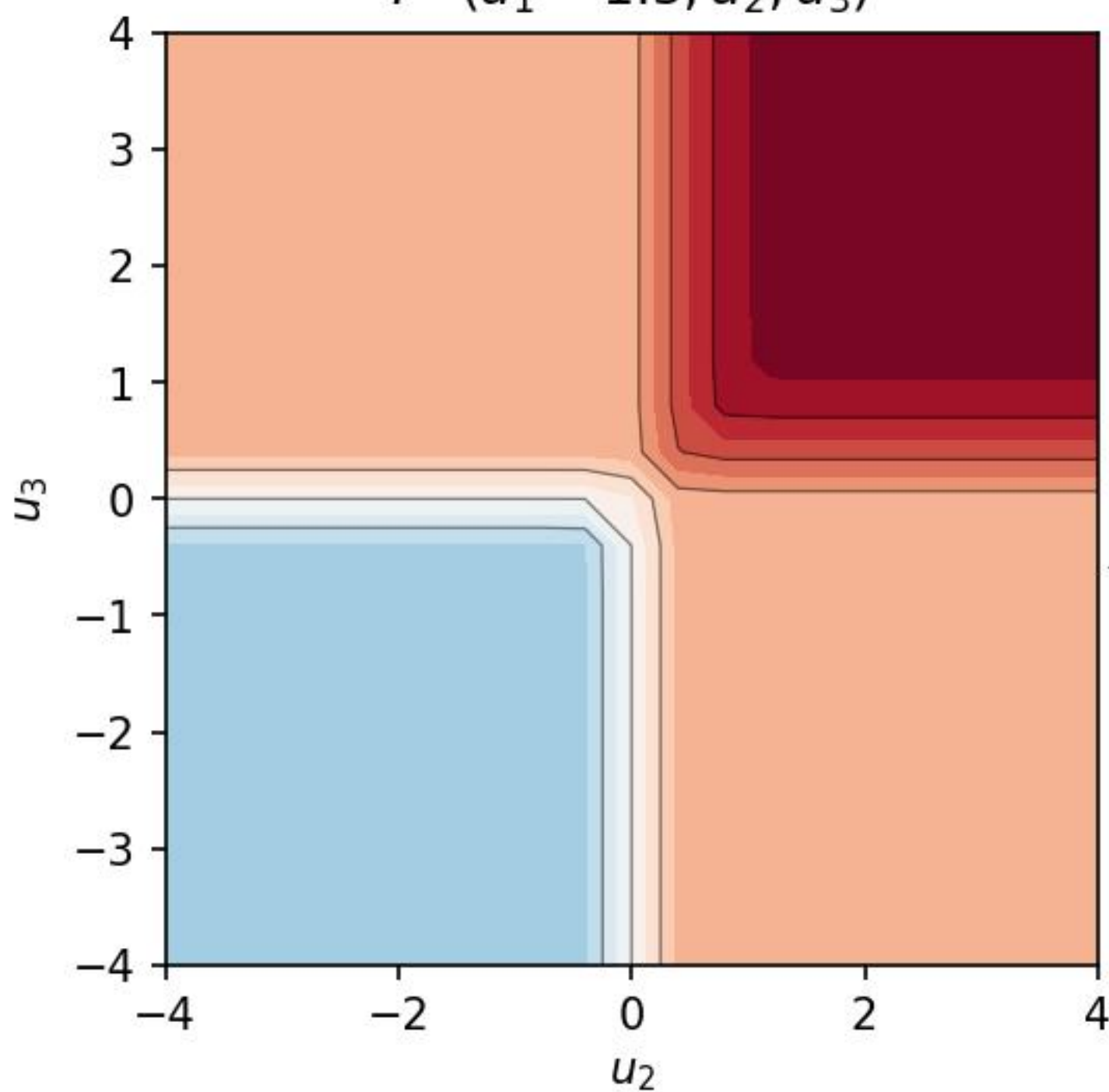
$P^*(u_1 = -1.5, u_2, u_3)$



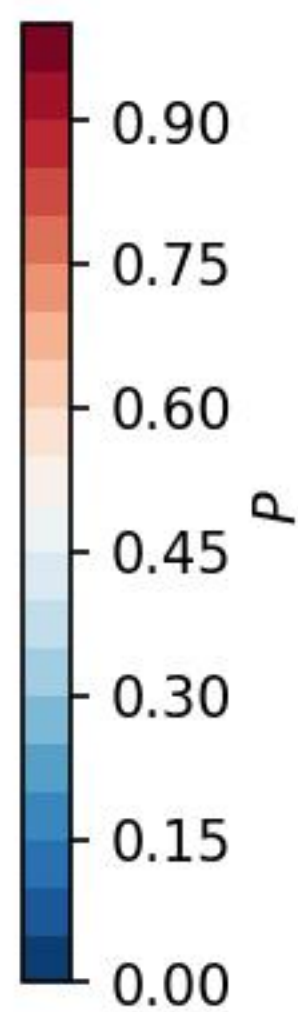
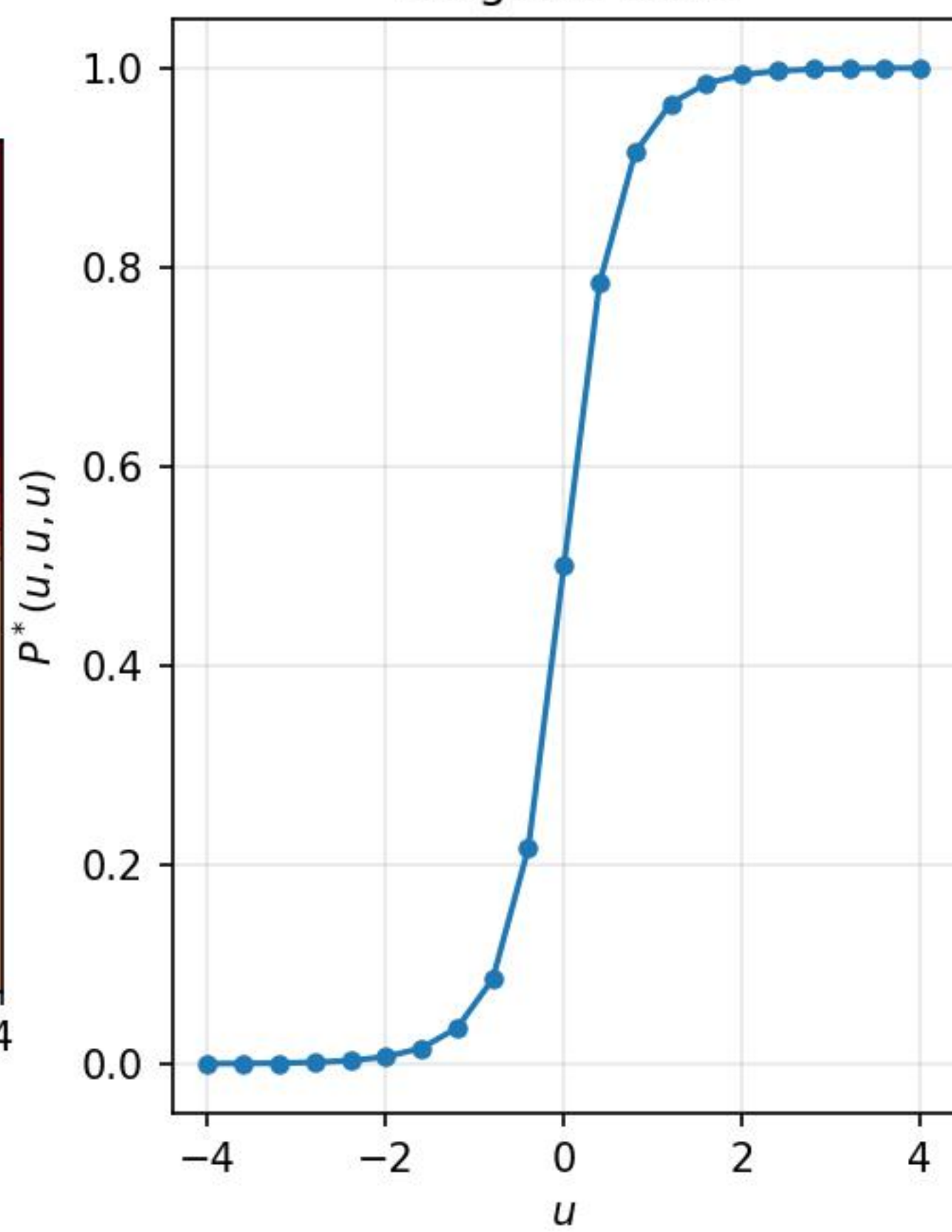
$P^*(u_1 = 0.0, u_2, u_3)$

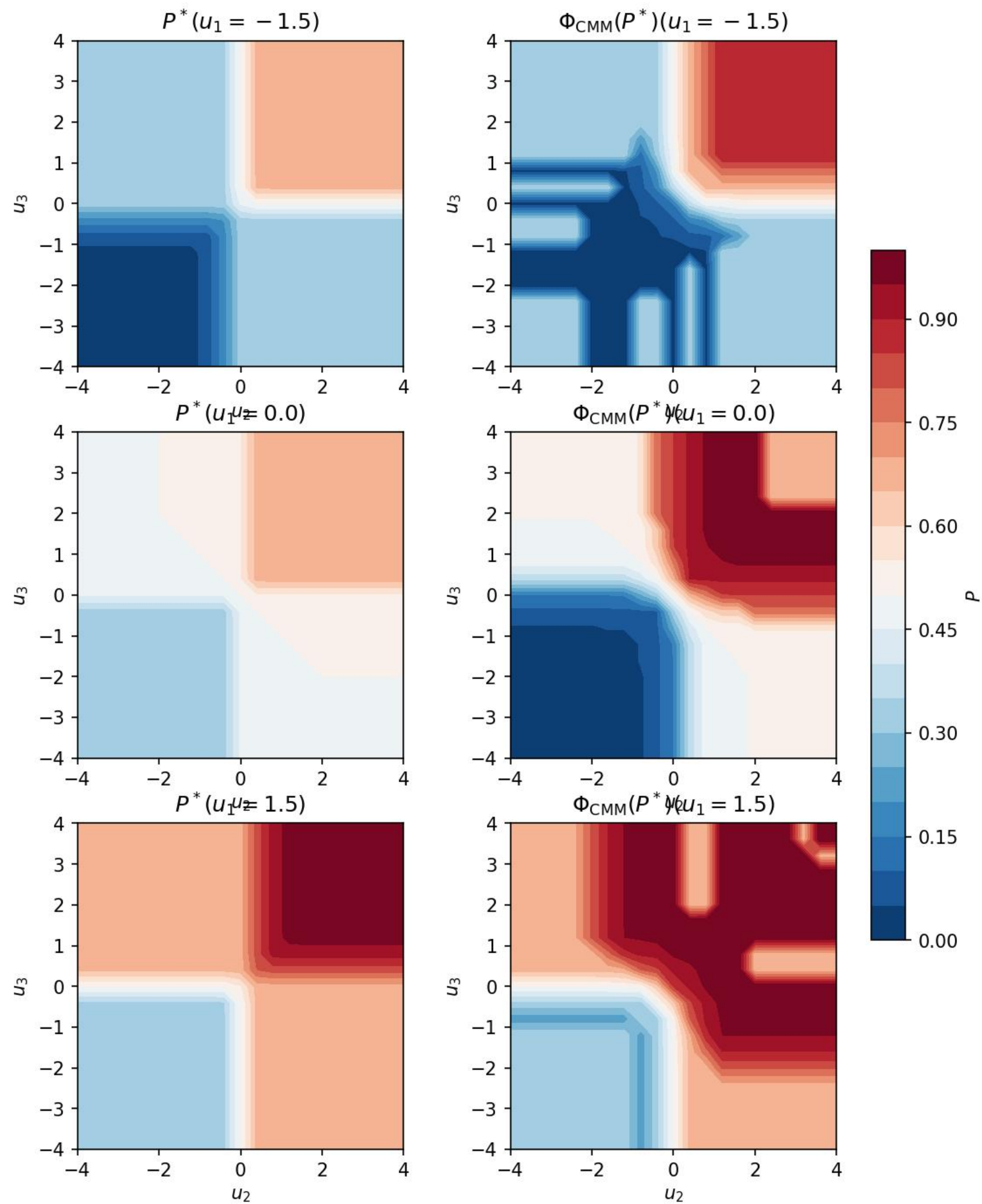


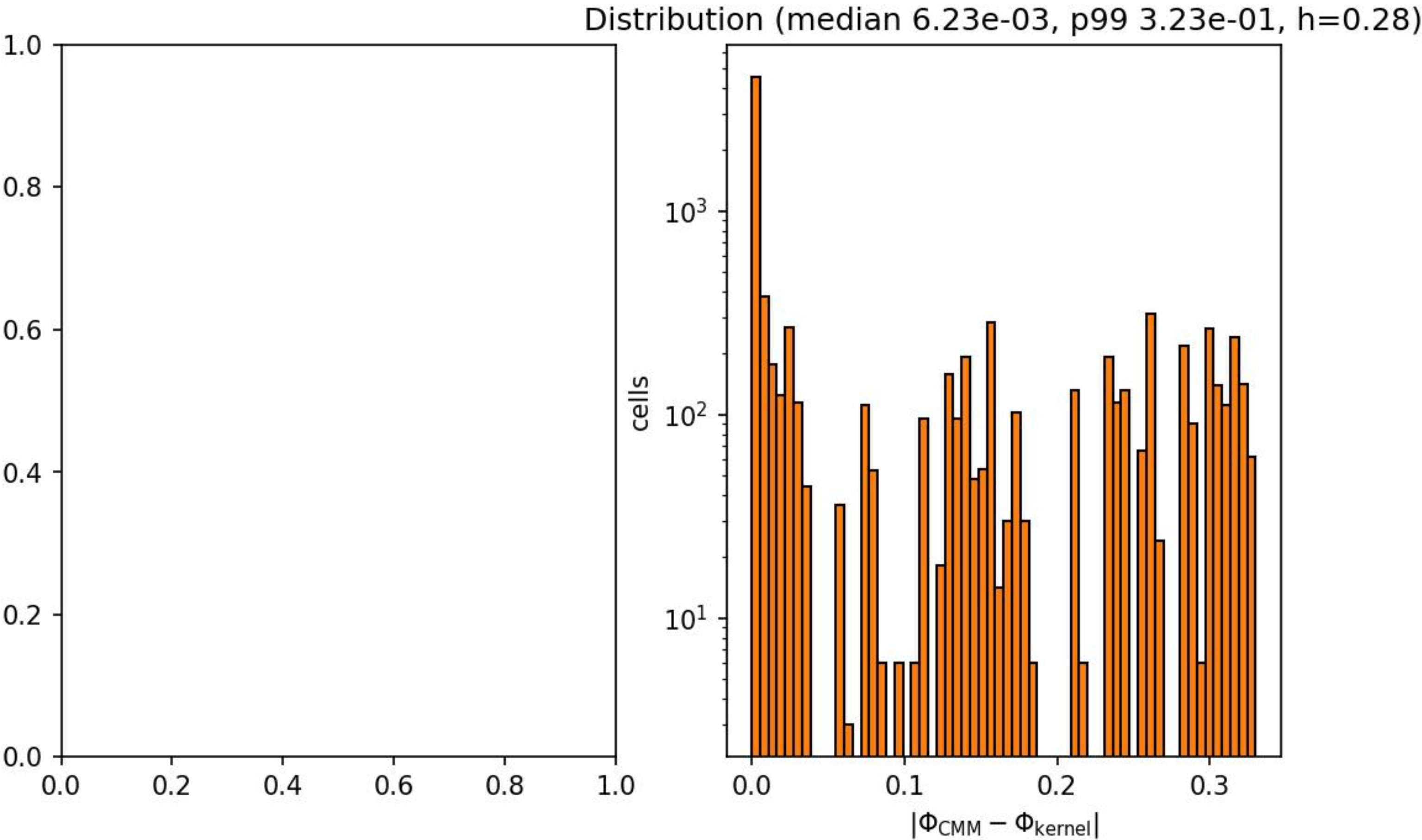
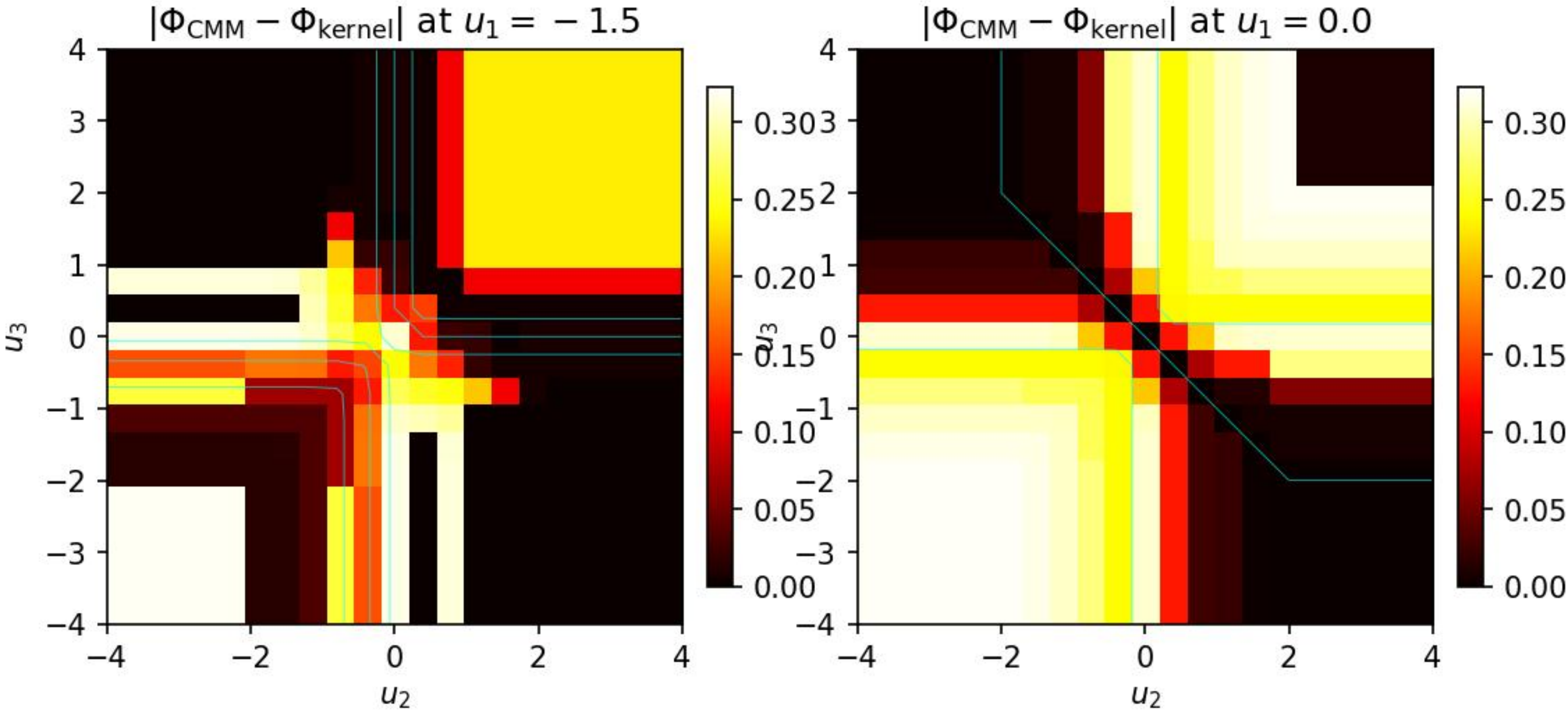
$P^*(u_1 = 1.5, u_2, u_3)$



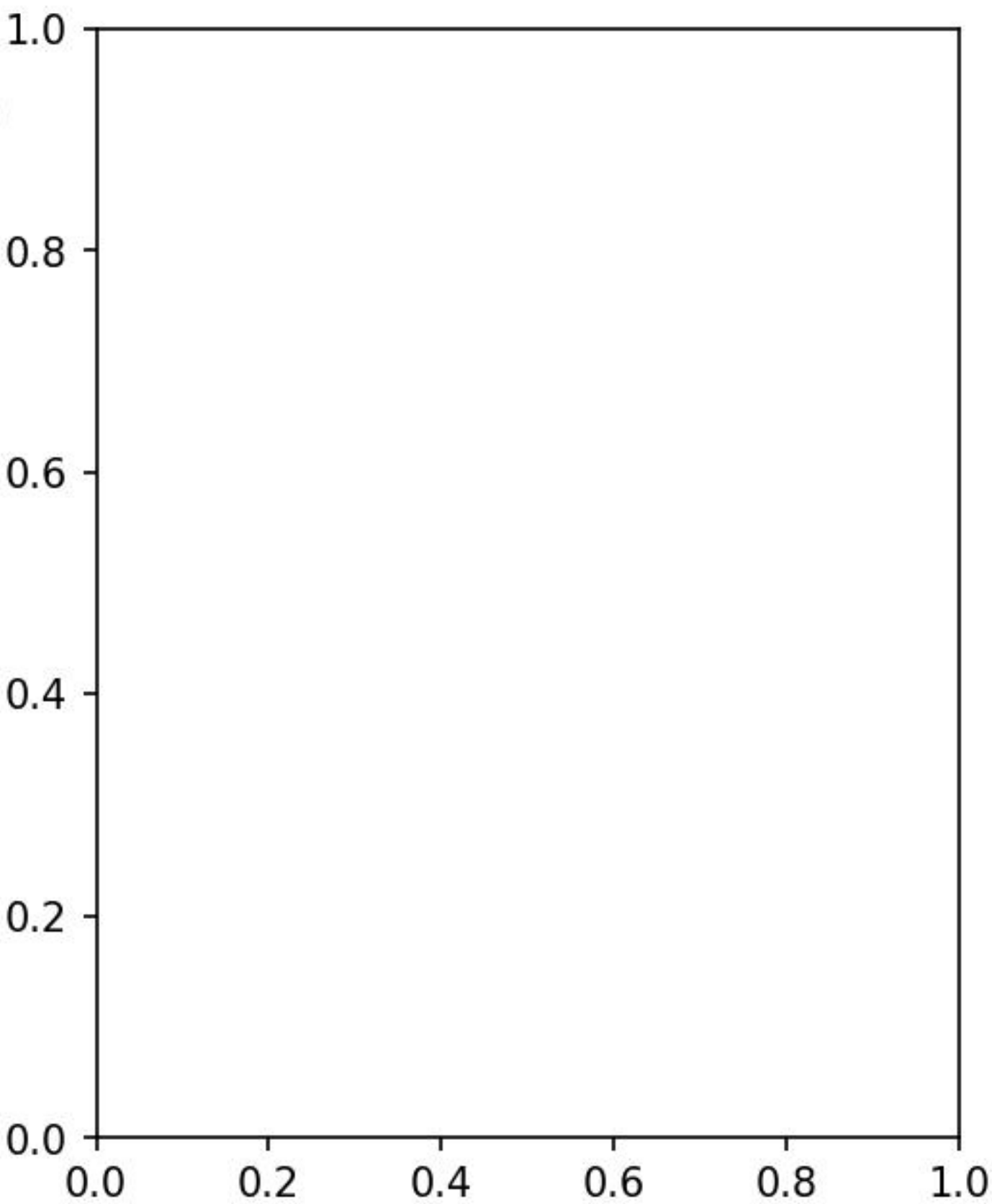
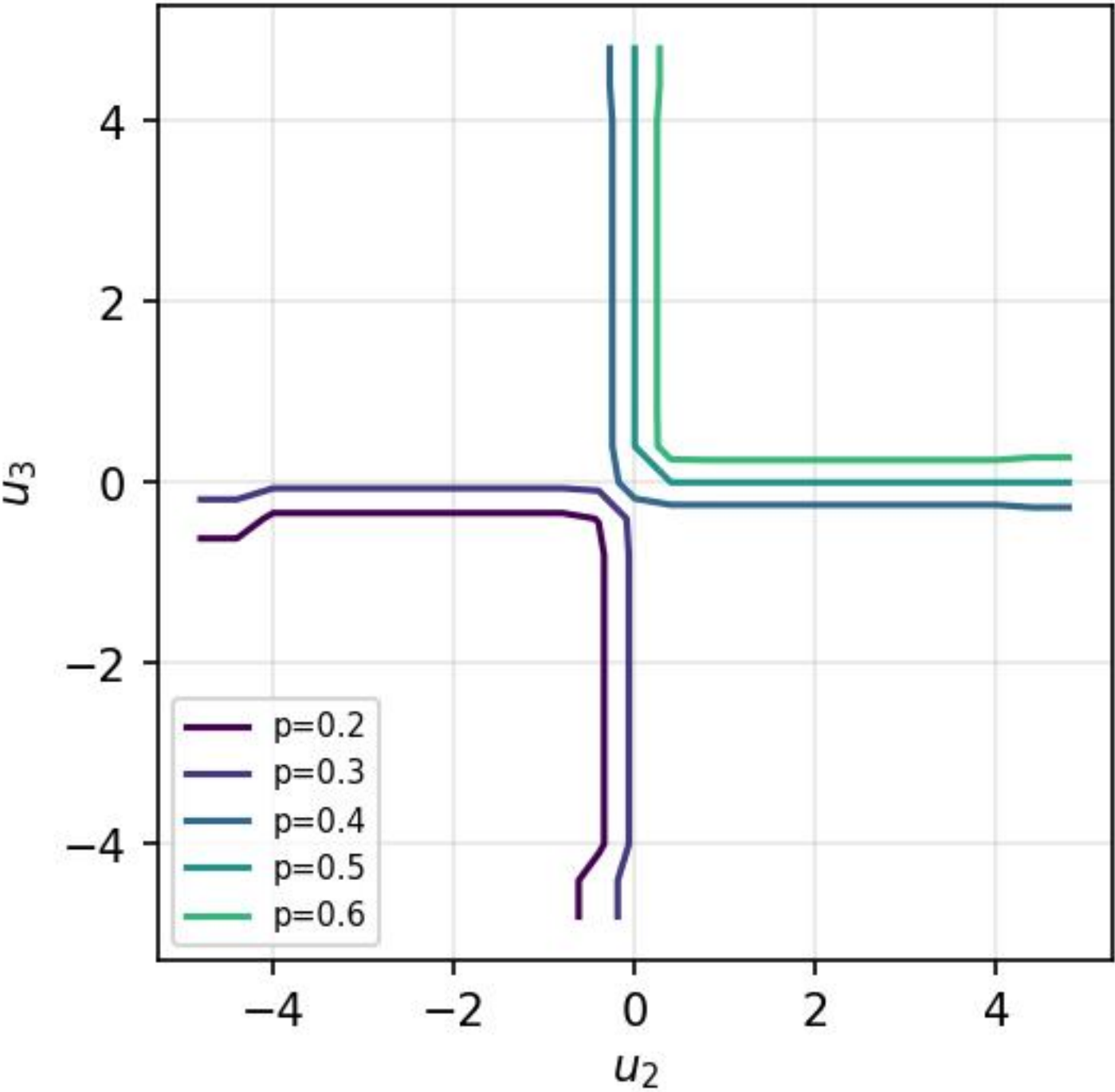
Diagonal slice







Level curves $\{P^* = p\}$ at $u_1 = -1.5$ (slice)



Selected level surfaces at $u_1 \in \{-0.5, 0.5, 1.0\}$

